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MTH - 645

(Mid-Term exam)

Q.No.1 :-

Set:- A set is collection of well defined objects.

e.g $A = \{1, 2, 3, 4, 5\}$ (set of No.)

$Z = \{0, 1, \dots\}$ (integers).

Q.No.2)

$$x \Delta_y x = ?$$

As we know that if x and y are two fuzzy set No. and Δ_y is t-norm then

$$x \Delta_y y = \frac{xy}{x+y-xy}$$

Now

$$x \Delta_y x = \frac{x \cdot x}{x+x-x \cdot x}$$

$$= \frac{x^2}{2x+x^2}$$

$$= \frac{x^2}{x(2+x)}$$

$$= \frac{x}{2+x} \quad \text{AR}$$

Q.No.3:-

Intersection of two intervals:-

Let $I_1 = [a_1, b_1]$

and $I_2 = [a_2, b_2]$ be two intervals
then their intersection is given as

$$I_1 \cap I_2 = [\max\{a_1, a_2\}, \min\{b_1, b_2\}]$$

and

$$I_1 \cap I_2 = \emptyset \text{ iff } a_1 > b_2 \text{ or } a_2 > b_1$$

Q.No.4

	x_1	x_2	x_3
q_1	0	1	0
q_2	1	0	0
q_3	0	1	0
q_4	0	0	1

write relation of R.

Sol:-

$$R = \{((q_1, x_1), 0), ((q_1, x_2), 1), ((q_1, x_3), 0), \\ ((q_2, x_1), 1), ((q_2, x_2), 0), ((q_2, x_3), 0), \\ ((q_3, x_1), 0), ((q_3, x_2), 1), ((q_3, x_3), 0), \\ ((q_4, x_1), 0), ((q_4, x_2), 0), ((q_4, x_3), 1)\}$$

Q.No.5:-

Prove that $(\tilde{A} \cap \tilde{A}') (x) \leq 0.5$

Proof:-

Consider $\tilde{A}$ be a fuzzy set
and $\tilde{A}(x)$ be its membership
value of x .

As by definition of complement
we have.

$$\tilde{A}'(x) = 1 - \tilde{A}(x) \rightarrow (1)$$

So

$$\begin{aligned} (\tilde{A} \cap \tilde{A}') (x) &= \min(\tilde{A}(x), \tilde{A}'(x)) \\ &= \min(\tilde{A}(x), 1 - \tilde{A}(x)) \end{aligned}$$

Case I)

$$\text{If } \tilde{A}(x) \geq 0.5$$

then

$$\tilde{A}'(x) = 1 - \tilde{A}(x) \leq 0.5$$

$$\Rightarrow \min(\tilde{A}(x), 1 - \tilde{A}(x)) \leq 0.5$$

Case II

$$\text{If } \tilde{A}(x) \leq 0.5$$

then

$$\tilde{A}'(x) \geq 0.5$$

$$1 - \tilde{A}(x) \geq 0.5$$

So

$$\min (\tilde{A}(x) \vee 1 - \tilde{A}(x)) \leq 0.5$$

So,

$$(\tilde{A} \cap \tilde{A}')(x) \leq 0.5 \text{ (proved)}$$

Q.No.6:-

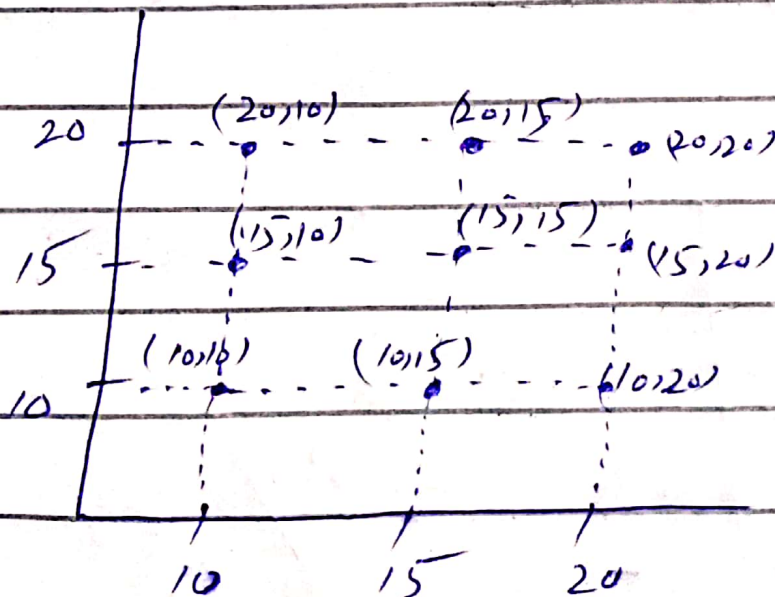
$A = \{10, 15, 20\}$ Find A^2 and show the graphically.

Sol:-

$$A = \{10, 15, 20\}$$

$$A \times A = \{10, 15, 20\} \times \{10, 15, 20\}$$

$$A^2 = \{ (10, 10), (10, 15), (10, 20), (15, 10), (15, 15), (15, 20), (20, 10), (20, 15), (20, 20) \}$$



Q.No.7:-

Support :- A set is said to be support of a fuzzy set $\tilde{A}$ if the membership value of the element is greater than 0.

i.e

$$\tilde{A} = \{(x_1, 0), (x_2, 0.1), (x_3, 0.4)\}$$

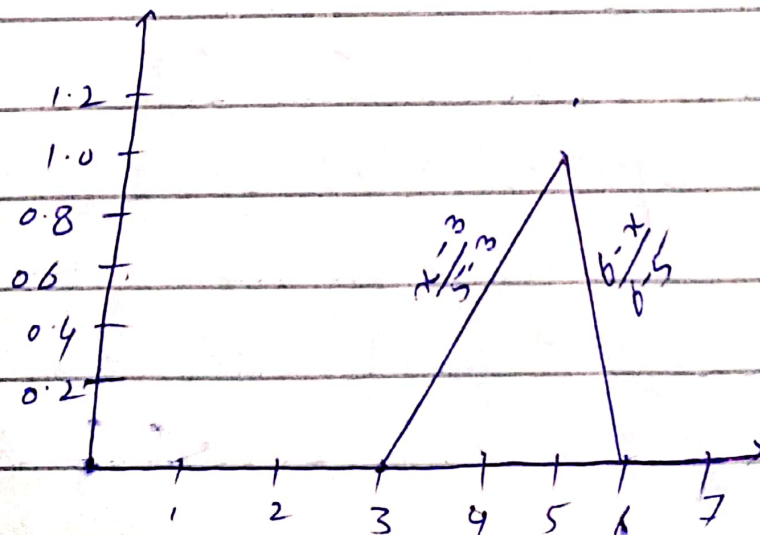
$$\text{Support of } \tilde{A} = \{x_2, x_3\}.$$

Q.No.8:-

Draw graph

$$\mu_A(x) = \begin{cases} \frac{x-3}{5-3} & 3 \leq x \leq 5 \\ \frac{6-x}{6-5} & 5 \leq x \leq 6 \\ 0 & \text{otherwise} \end{cases}$$

sol:-



Q.No.9

$$\mu_A = \begin{cases} \frac{x-a}{b-a} & a \leq x \leq b \\ \frac{c-x}{c-b} & b \leq x \leq c \\ 0 & \text{otherwise} \end{cases}$$

Find α_1 and α_2 and
written as $\mu_x = [\alpha_1, \alpha_2]$

Sol:-

$$\alpha = \frac{x-a}{b-a}$$

$$\alpha(b-a) = x-a$$

$$\alpha(b-a) + a = x$$

$$\alpha_1 = x = \alpha(b-a) + a$$

and

$$\alpha = \frac{c-x}{c-b}$$

$$\alpha(c-b) = c-x$$

$$-x = \alpha(c-b) + x$$

$$\alpha_2 = x = c - \alpha(c-b)$$

Now

$$A_x = [x_1, x_2]$$

$$= [x(b-a) + a, c - x(c-b)]$$

Q.No. 10):-

Make the membership table
for fuzzy relation

$$R = \{ (a, c), 0.7, (b, a), 0.9, \\ (c, b), 1, (c, d), 0.2 \}$$

Sol:-

M_R	a	b	c	d
a	0	0	0.7	0
b	0.9	0	0	0
c	0	1	0	0.2

Q.No. 11):-

find $0.8 \Delta_2 0.7$

As we know that if

x and y are two fuzzy
No. then

$$x \Delta_2 y = \frac{xy}{2 - (x + y - xy)}$$

So

$$\begin{aligned} 0.8 \Delta_2 0.7 &= \frac{(0.8)(0.7)}{2 - (0.8 + 0.7 - (0.8)(0.7))} \\ &= \frac{0.56}{2 - (1.5 - 0.56)} \\ &= \frac{0.56}{2 - 0.94} \\ &= \frac{0.56}{1.06} \\ &= 0.53 \end{aligned}$$

Malik

Q.No.12)

$$x \Delta_1 x$$

Ans

$$x \Delta_1 y = OV(x + y - 1)$$

So

$$\begin{aligned} x \Delta_1 x &= OV(x + x - 1) \\ &= OV(2x - 1) \end{aligned}$$

Q.No.13)

$$x + (y + z) = (x + y) + z$$

Finally

$$\begin{aligned} x + (y + z) &= \{x + a \mid x \in X, a \in y + z\} \\ &= \{x + (y + z) \mid x \in X, y \in Y, z \in Z\} \end{aligned}$$

$$= \{(x+y)+z \mid x \in X, y \in Y, z \in Z\}$$

$$= \{b+z \mid b \in X+Y, z \in Z\}$$

$$= (X+Y)+Z$$

(Q.No. 14)

M_R	a	b	c
1	0.1	1.0	0.0
2	0.3	1.0	1.0
3	0.0	1.0	1.0

Find Fuzzy Relation.

Sol.:-

$$R = \{(1, a), 0.1), (1, b), 1.0), (1, c), 0.0), \\ (2, a), 0.3), (2, b), 1.0), (2, c), 1.0), \\ (3, a), 0.0), (3, b), 1.0), (3, c), 1.0)\}.$$

Q.No. 15)

Union of two fuzzy set:-
Let $\tilde{F}(x)$ be

the collection of fuzzy sets
and $\tilde{A}, \tilde{B} \in \tilde{F}(x)$ then

the Union of $\tilde{A}$ and $\tilde{B}$
is the fuzzy set $\tilde{C}$ which
is defined as

$$\mu_{\tilde{A} \cup \tilde{B}} = \max \{ \mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x) \}$$

$$= \mu_{\tilde{A}}(x) \vee \mu_{\tilde{B}}(x), \forall x \in X$$

Q.No. 16

Degree of subsethood:-

Degree of subsethood
of $\tilde{A}$ in $\tilde{B}$ is defined as

$$S(\tilde{A}, \tilde{B}) = \frac{|\tilde{A} \cap \tilde{B}|}{|\tilde{A}|}$$